CO4 AT3

COMPARATIVE ANALYSIS

1. Object Detection Performance Comparison

Given: Model A has 80 True Positives (TP), 20 False Positives (FP), and 30 False Negatives (FN). Model B has 90 True Positives (TP), 15 False Positives (FP), and 20 False Negatives (FN).

(a) Calculate the Precision and Recall for both models.

Precision measures the proportion of detected objects that are actually correct. Recall measures the proportion of actual objects that are correctly detected.

Precision = TP / (TP + FP) × 100

Recall = TP / (TP + FN) × 100

Model A:

Precision = 80 / (80 + 20) × 100 = 80%

Recall = 80 / (80 + 30) × 100 = 72.73%

Model B:

Precision = 90 / (90 + 15) × 100 = 85.71%

Recall = 90 / (90 + 20) × 100 = 81.82%

(b) Comparison of Results

Model	Precision	Recall
Model A	80%	72.73%
Model B	85.71%	81.82%

(c) Justification

Model B provides better overall object detection performance because it achieves higher precision and recall than Model A. The precision of Model B is 85.71%, compared with 80% for Model A, indicating that Model B produces fewer false positive detections. Its recall is 81.82%, compared with 72.73% for Model A, indicating that Model B detects a larger proportion of the actual objects. Therefore, Model B is more reliable for object detection applications.

Final Answer: Model B provides better overall object detection performance.

2. Motion Estimation Error Analysis

Given: Method A produces displacement errors of 2, 3, and 2 pixels. Method B produces displacement errors of 3, 5, and 4 pixels for three consecutive frames.

(a) Calculate the Average Displacement Error.

Average Displacement Error = Sum of Displacement Errors / Number of Frames

Method A:

Average Error = (2 + 3 + 2) / 3 = 7 / 3 = 2.33 pixels

Method B:

Average Error = (3 + 5 + 4) / 3 = 12 / 3 = 4 pixels

(b) Comparison

Method	Frame Errors	Average Error
Method A	2, 3, 2 pixels	2.33 pixels
Method B	3, 5, 4 pixels	4.00 pixels

(c) Justification

Method A is more accurate because it has a lower average displacement error. Its average error is 2.33 pixels, whereas Method B has an average error of 4 pixels. The lower error means that the estimated motion is closer to the actual motion. Method A also shows more consistent errors across the three frames, while Method B produces larger errors in every frame.

Final Answer: Method A provides more accurate and reliable motion estimation.

3. Depth Estimation Comparison

Given: The actual depths of three objects are 2 m, 3 m, and 4 m. Stereo Vision estimates them as 2.0 m, 3.0 m, and 4.0 m. Structure-from-Motion (SfM) estimates them as 2.5 m, 3.8 m, and 5.0 m.

Object	Actual Depth	Stereo Vision	SfM
1	2.0 m	2.0 m	2.5 m
2	3.0 m	3.0 m	3.8 m
3	4.0 m	4.0 m	5.0 m

(a) Calculate the Absolute Error for Each Object.

Absolute Error = |Estimated Depth − Actual Depth|

Stereo Vision:

Object 1: $|2.0 - 2.0| = 0$ m

Object 2: $|3.0 - 3.0| = 0$ m

Object 3: $|4.0 - 4.0| = 0$ m

Structure-from-Motion (SfM):

Object 1: $|2.5 - 2.0| = 0.5$ m

Object 2: $|3.8 - 3.0| = 0.8$ m

Object 3: $|5.0 - 4.0| = 1.0$ m

(b) Calculate the Average Absolute Error.

Stereo Vision:

Average Absolute Error = (0 + 0 + 0) / 3 = 0 m

SfM:

Average Absolute Error = (0.5 + 0.8 + 1.0) / 3 = 2.3 / 3 = 0.77 m

(c) Comparison and Evaluation

Method	Object 1	Object 2	Object 3	Average Error
Stereo Vision	0 m	0 m	0 m	0 m
SfM	0.5 m	0.8 m	1.0 m	0.77 m

Stereo Vision provides the most accurate depth estimation because its estimated depths exactly match the actual depths for all three objects. Its average absolute error is 0 m, while SfM has an average absolute error of 0.77 m. The errors in the SfM estimates also increase from 0.5 m to 1.0 m across the three objects.

Therefore, Stereo Vision is more accurate for the given data because it has the lowest possible average error.